

David Liu

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EDUCATION

University of Western Ontario

B.E.Sc. in Software Engineering

London, ON

Sep 2022 – Apr 2028 (Expected)

Awards: Winner, Best Use of Solana and Badge Hack (\$2,500) – Hack the North 2026; 2nd Place (\$5,000) – TELUS AI at the Edge of Innovation; Top 5 Finalist – UofT Hackathon; Finalist – Western Engineering Competition

Involvements: AI Research Assistant – Tsinghua University (2023), Western Founders Network (VP of Education), Hack Western Mentor, Western AI Mentor, Ignition Hacks Judge, Muay Thai Club Beginner-Class Coach, Classical Guitarist (Carnegie Hall)

PROFESSIONAL EXPERIENCE

J.D. Power

Software Engineering Intern (16-month Co-op)

London, ON

May 2025 – Aug 2026

- Built and maintained **Spring Boot** services on **Kubernetes** powering Build & Price sites and analytics connectors for five Stellantis brands (**Jeep, RAM, Dodge, Chrysler, Maserati**), used by millions of shoppers
- Fixed backend and data-pipeline defects blocking the model-year 2026 site launch, restoring the customer vehicle-customization flow and shipping accurate configurations to production
- Remediated **2,000+** TruffleHog secret-scanning findings across automotive codebases, rotating leaked keys, and authored a research report on **GitHub Copilot** for automated vulnerability detection

Tethos Association

Founder and President

London, ON

May 2024 – Current

- Solo-founded Western's largest student-run tech organization, now a federally incorporated non-profit building pro-bono software for other non-profits: **250+** student developers, an **80+** executive team, and **2** chapters (Western, UBC) in 2 years
- Delivered **20+** pro-bono software projects for non-profits including **World Vision**, the **Canadian Red Cross**, and the **London Children's Museum**, saving partners **\$500,000+** in development costs
- Produced **GENESIS**, London's largest student demo day: **260+** attendees, **7** project demos, **8** sponsors, and **\$12,000** in sponsorships

Modern Engineering

Software & Robotics Engineering Intern

Delta, BC

May 2024 – Sep 2024

- Programmed a **UR5** collaborative robot arm in **Python** over the **RTDE** interface: a **6**-program motion library spanning object retrieval, conveyor placement, part sanding, and **OpenCV** camera-guided stacking

PERSONAL PROJECTS

tethos.ca – Tethos Association Platform

Next.js, TypeScript, FastAPI, Supabase, React Three Fiber | GitHub

May 2026 – Current

- Solo-engineered a **2.5D** MMO game world as the club's member portal, with real-time multiplayer via **Colyseus**: **875+** 3D assets, procedural terrain, a **91**-species fishing system, and a dual-currency economy enforced by server-side **Postgres** RPCs
- Created **5** LLM NPCs (**GPT-6 Luna** + **FastAPI RAG**) persona-cloned from real exec members, with per-player memory files reloaded before each chat so characters remember members, build relationships, and evolve
- Architected an autonomous multi-agent dev pipeline: role-scoped **Claude Code** agents (build, QA, reviewer) with spec-driven sprints and QA gates, shipping **70+** unattended build iterations
- Built the surrounding production platform for the **400+** member non-profit: public site, **59** API routes, **5**-tier role-based access, and a recruitment kanban dashboard that has processed **240+** applications

Among Us IRL – Winner, Best Use of Solana and Badge Hack (\$2,500)

C++, ESP32-C3, ESP-NOW, PlatformIO | Hack the North | Devpost

Sep 2026

- Built the game software for real-life Among Us on **ESP32-C3** badges in **C++**: role dealing, the kill/report/emergency-meeting state machine, on-badge voting with live tallies, and an **NFC** tag-scanning task system with **IMU**-driven minigames
- Runs with no server, router, or phones: game state syncs peer-to-peer over an **ESP-NOW** TTL-flood mesh, with **RSSI** signal strength gating kills and body reports to real physical proximity

TECHNICAL SKILLS

Languages: Python, Java, C++, Go, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Express, Spring Boot, FastAPI, Flask, React Three Fiber, GSAP, Framer Motion, Tailwind

AI & Data: OpenAI & Anthropic APIs, RAG, OpenCV, TensorFlow, PyTorch, PostgreSQL, Supabase, MySQL, MongoDB, Kafka

Systems & Tools: ESP32/Arduino/PlatformIO, Docker, Kubernetes, AWS, Linux, Git, GitHub Actions (CI/CD), REST APIs